

Problem 4

$$C_{\text{ppm}} = \frac{Q_m R_g T}{k Q_v P M} \cdot 10^6$$

$$Q_m = \frac{C_{\text{ppm}} k Q_v P M}{R_g T} \cdot 10^{-6}$$

$$C_{\text{ppm}} = 10$$

$$P = 1 \text{ atm}$$

$$T = 80^\circ\text{F} = 540^\circ\text{R}$$

$$Q_v = 2000 \text{ ft}^3/\text{min}$$

$$R_g = 0.7302 \text{ ft}^3 \cdot \text{atm}/\text{lb-mole}/^\circ\text{R}$$

$$k = \frac{1}{8}$$

$$M = 34 \text{ lb-mole}/\text{lb}$$

$$Q_m = \frac{10 \cdot \frac{1}{8} \cdot \frac{2000}{60} \cdot 1 \cdot 34}{0.7302 \cdot 540} \cdot 10^{-6}$$

$$Q_m = 3.59 \cdot 10^{-6} \text{ lb/s}$$

$$Q_m = C_0 A P_0 \sqrt{\frac{\gamma g_c M}{R_g T_0} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}}}$$

$$A = \frac{Q_m}{C_0 P_0 \sqrt{\frac{\gamma g_c M}{R_g T_0} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}}}$$

$$C_0 = 1$$

$$P_0 = 100 \text{ psig} = 114.7 \text{ psig}$$

$$\gamma = 1.3$$

$$R_g = 10.73 \text{ ft}^3 \cdot \text{atm}/\text{lb-mole}/^\circ\text{R}$$

$$g_c = 31.174 \text{ lb}_m \cdot \text{ft}/\text{lb}_f/\text{s}^2$$

$$A = \frac{3.59 \cdot 10^{-6}}{1 \cdot 114.7 \cdot \sqrt{\frac{1.3 \cdot 32.174 \cdot 34}{144 \cdot 10.73 \cdot 540} \left(\frac{2}{1.3+1} \right)^{\frac{1.3+1}{1.3-1}}}}$$

$$A = 1.29 \cdot 10^{-6} \text{ in}^2$$

$$A = \frac{1}{4} \pi D^2$$

$$D = \sqrt{\frac{4A}{\pi}}$$

$$D = \sqrt{\frac{4 \cdot 1.29 \cdot 10^{-6}}{\pi}}$$

$$\boxed{D = 0.00128 \text{ in}}$$